

JEFF KIM

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TECHNICAL SKILLS

Languages:

- C, C++, Java, JavaScript, TypeScript, Python, MySQL, HTML5, PHP

Frameworks & Tools:

- React, jQuery, OpenGL, Oracle, Git, CMake, JUnit, Mocha

Programming Concepts:

- Machine Learning & Model Training
- REST API Development, Relational Databases
- Database Analysis & Optimization
- Network Programming (TCP/IP, Sockets, Encryption)
- Threading & Concurrency (POSIX Threads)
- Unit / Integration / End-to-End Testing
- Test-Driven Development (TDD)

TECHNICAL PROJECTS

Game Development using C++, OpenGL, and JIRA (Academic/Personal) January 2024 – Present

- Designed a third-person shooting game in C++ with OpenGL, GLFW, SDL, and CMake, simulating real-time interactions between agents and events.
- Engineered a custom physics and collision detection system using AABB and SAT for entity interactions, supporting precise control logic and responsiveness.
- Applied entity component system (ECS) to support scalability, maintainability, and code reuse across game entities and systems.
- Developed multiple flexible systems to manage enemy AI, user input, and UI feedback.
- Integrated real-time visual feedback using dynamic lighting and font rendering, improving user engagement and interface clarity.
- Emphasized testable systems, memory management, and performance optimization
- Utilized Jira to apply SCRUM methodologies, including sprint planning, task breakdown, weekly meetings, and retrospectives to manage team progress and ensure iterative development.

Relational Database Design for Social Media Platform (Academic) Sept 2025 – Dec 2025

- Designed and implemented a relational database schema for a social media platform featuring users, communities, posts, comments, messaging, and awards.
- Modeled complex user interactions and entity relationships using ER diagrams and normalized schemas to support scalability and data integrity.
- Implemented SQL operations to handle core user actions such as account creation, content interaction, and data retrieval.
- Developed advanced SQL queries using 'join' and 'aggregation' to support key features like award search, post engagement tracking, and user-community associations.
- Focused on creating efficient and maintainable database structures that reflect real-world social networking behaviors

Java Dictionary Client for DICT.org Protocol (Academic) January 2025 – March 2025

- Built a Java GUI client interfacing with DICT.org over TCP.
- Implemented core networking functionality in application to connect a Java client to the DICT.org dictionary server using a custom protocol.
- Handled command parsing, socket communication, and response interpretation to enable word lookup, multi-database search, and auto-complete strategies.

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- Worked with dropdown-based GUI components to support dynamic database and matching strategy selection.
- Built and tested the application for reliable compilation and deployment, emphasizing user-friendly functionality and a clean, intuitive interface.

Predicting Features using Machine Learning Algorithms (Personal)

Sept 2023-Sept 2025

- Preprocessed data with scikit-learn pipelines and engineered features for model inputs.
- Trained ML models (linear, tree-based, ensemble) using PyTorch.
- Applied randomized search for tuning and SHAP for analysis of feature relationships.

WORK EXPERIENCE

Chef (Canuel Caterers)

Sept 2021 – June 2023

Ice Water Café, Coquitlam, BC

- Prepared over 200 meals daily in a fast-paced environment by staying organized, and focused.
- Trained over 30 new staff members emphasizing teamwork and proper food handling and procedures to ensure food safety and good teamwork.

Lifeguard and Swimming Instructor

Sept 2016 – Feb 2019

University of Manitoba, Winnipeg, MB

- Planned beginner to advanced swimming level lesson plans according to each students strengths and weakness using different teaching methods and activities to keep students interested.

EDUCATION

Bachelor of Computer Science (BCS)

Sept 2022 – April 2025

University of British Columbia, Faculty of Science, Vancouver, BC

Bachelor of Science (B.Sc.)

Sept 2018 – May 2022

University of British Columbia, Faculty of Science, Vancouver, BC

- Combined Major in Chemistry, Life Science, and Earth and Environmental Science